Mount Kenya University

Unlocking Infinite Possibilities

BIT 4131 Statistical Computing and Reporting: Group Work

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Abstract

This paper summarises the theory and mathematical formulation of 23 statistical computing topics, from descriptive statistics and probability to regression, multivariate analysis, resampling, missing data and reproducible reporting. Each section states the method, its key formulas and a short hand-worked example, and reports a matching result from our Python implementation. All data are synthetic and generated with fixed random seeds, so every result in the repository can be reproduced exactly. The full code, datasets, output tables, plots and interpretation for each topic are in the linked GitHub folders.

Keywords: statistical computing, Python, simulation, inference, regression, reproducibility.
Repository: https://github.com/Isaac-Y67/statistical-computing

1 Measures of central tendency and dispersion

Theory. Measures of central tendency describe where the data are centred. Measures of dispersion describe how spread out the data are. Quartiles split the ordered data into four equal parts.

Formulation.

Mean	$\bar{x} = (1/n)\,\Sigma\, x_i$
Median	Middle value of the ordered data (average of the two middle values if n is even)
Mode	Most frequent value
Sample variance	$s^2 = \Sigma(x_i - \bar{x})^2 \,/\, (n-1)$
Standard deviation	$s = \sqrt{s^2}$
Range and IQR	$\text{Range} = \max - \min;\ \ \text{IQR} = Q3 - Q1$
Outlier fences	$\text{Lower} = Q1 - 1.5 \times \text{IQR};\ \ \text{Upper} = Q3 + 1.5 \times \text{IQR}$

Worked example.

Given	**Data: 4, 8, 6, 5, 3**
Mean	$(4 + 8 + 6 + 5 + 3) \,/\, 5 = 26 \,/\, 5 = 5.2$.
Median	the ordered data are 3, 4, 5, 6, 8, so the median is 5.
Deviations from the mean	$-1.2, 2.8, 0.8, -0.2, -2.2$. Squared: 1.44, 7.84, 0.64, 0.04, 4.84. Sum = 14.8.
Variance	$s^2 = 14.8 \,/\, (5-1) = 3.7$. Standard deviation: $s = \sqrt{3.7} = 1.924$.
Coefficient of variation	$1.924 \,/\, 5.2 \times 100 = 37.0\%$.

Result in our analysis. In the exam-scores analysis, math has a mean of 61.36 and a median of 61.25 (symmetric), and science has the largest CV (32.12%).

Code and results: topic_01_central_tendency_dispersion

2 Probability and conditional probability

Theory. Probability measures how likely an event is, on a scale from 0 (impossible) to 1 (certain). Conditional probability P(A | B) is the probability of A once we know that B has occurred. Bayes' theorem reverses the direction of conditioning: from P(B | A) we can obtain P(A | B).

Formulation.

Probability	$P(A) = \text{number of favourable outcomes} \,/\, \text{total outcomes},\ \ 0 \le P(A) \le 1$
Complement	$P(A^c) = 1 - P(A)$
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional probability	$P(A \mid B) = P(A \cap B) \,/\, P(B),\ \ \text{for } P(B) > 0$
Multiplication rule	$P(A \cap B) = P(A \mid B) \cdot P(B)$
Independence	$P(A \mid B) = P(A)$, equivalently $P(A \cap B) = P(A) \cdot P(B)$
Total probability	$P(B) = \Sigma\, P(B \mid A_i) \cdot P(A_i)$

Worked example.

Given	**Customer churn and support calls (from the 1,000-customer dataset)**
From the data	$P(\text{Call}) = 0.284$, $P(\text{Churn}) = 0.191$ and $P(\text{Call and Churn}) = 0.112$.
Conditional probability	$P(\text{Churn} \mid \text{Call}) = 0.112 / 0.284 = 0.394$.
Independence check	$P(\text{Churn}) = 0.191$ differs greatly from 0.394, so churn and support calls are not independent.
Bayes' theorem	$P(\text{Call} \mid \text{Churn}) = P(\text{Churn} \mid \text{Call}) \times P(\text{Call}) / P(\text{Churn}) = 0.394 \times 0.284 / 0.191 = 0.586$.

Result in our analysis. The Python output gives P(Churn | Call) = 0.3944 and P(Call | Churn) = 0.5864, both by Bayes and directly from the data.

Code and results: topic_02_probability

3 Probability distributions

Theory. A random variable assigns a number to the outcome of a random process. Discrete variables use a probability mass function (PMF), continuous variables use a probability density function (PDF), and both have a cumulative distribution function (CDF). Parameters are estimated from data by maximum likelihood, which chooses the values that make the observed data most probable.

Formulation.

CDF	$F(x) = P(X \leq x)$
Normal(μ, σ^2)	$f(x) = (1 / (\sigma\sqrt{(2\pi)})) \cdot \exp(-(x-\mu)^2 / (2\sigma^2))$; mean μ, variance σ^2; MLE: $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = \Sigma(x_i - \bar{x})^2 / n$
Standardisation	$Z = (X - \mu) / \sigma \sim N(0, 1)$; $P(X < a) = \Phi((a - \mu) / \sigma)$
Binomial(n, p)	$P(X = k) = C(n, k)\, p^k\, (1-p)^{n-k}$; mean np, variance $np(1-p)$; MLE: $\hat{p} = \bar{x} / n$
Poisson(λ)	$P(X = k) = e^{-\lambda}\, \lambda^k / k!$; mean = variance = λ; MLE: $\hat{\lambda} = \bar{x}$
Exponential(λ)	$f(x) = \lambda\, e^{-\lambda x}$ for $x \geq 0$; mean $1/\lambda$, variance $1/\lambda^2$; $P(T > t) = e^{-\lambda t}$; MLE: $\hat{\lambda} = 1 / \bar{x}$
Kolmogorov-Smirnov	$D = \max \lvert F_n(x) - F(x) \rvert$; H_0: the data follow the fitted distribution

Worked example.

Given	**Using the fitted parameters from the analysis**
Normal	height has mean 169.893 and sd 7.671. For $P(H < 165)$: $Z = (165 - 169.893) / 7.671 = -0.638$, so $P = \Phi(-0.638) \approx 0.262$.
Binomial	defects in a batch of 20 with $p = 0.1013$. $P(X = 0) = (1 - 0.1013)^{20} = 0.8987^{20} \approx 0.118$.
Poisson	arrivals per hour with $\lambda = 4$. $P(X = 0) = e^{-4} \approx 0.0183$.
Exponential	waiting time with rate $\lambda = 0.2209$ per minute. $P(T > 10) = e\text{^}(-0.2209 \times 10) = e\text{^}(-2.209) \approx 0.110$.

Result in our analysis. The Python results are 0.2618, 0.1181, 0.0183 and 0.1099, matching the hand calculations.

Code and results: topic_03_probability_distributions

4 Sampling distributions and the Central Limit Theorem

Theory. A statistic such as the sample mean changes from one random sample to the next. The sample mean is unbiased: its sampling distribution is centred on the population mean. The Central Limit Theorem (CLT) states that, for a population with finite variance, the sampling distribution of the mean approaches a normal distribution as n grows, whatever the shape of the population.

Formulation.

Mean of the sample mean	$E(\bar{X}) = \mu$

Variance of the sample mean	$\mathrm{Var}(\bar{X}) = \sigma^2 / n$
Standard error	$\mathrm{SE}(\bar{X}) = \sigma / \sqrt{n}$; estimated from a sample by $s / \sqrt{n}$
Central Limit Theorem	$(\bar{X} - \mu) / (\sigma / \sqrt{n}) \to N(0, 1)$ as $n \to \infty$
Standard error of a proportion	$\mathrm{SE}(\hat{p}) = \sqrt{(p(1-p) / n)}$
Normal approximation	$P(\bar{X} > c) \approx 1 - \Phi((c - \mu) / (\sigma / \sqrt{n}))$

Worked example.

Given	**Right-skewed population with $\mu = 10.035$ and $\sigma = 10.053$**
Standard error for n = 30	$\mathrm{SE} = 10.053 / \sqrt{30} = 10.053 / 5.477 = 1.835$.
Standard error for n = 100	$\mathrm{SE} = 10.053 / 10 = 1.005$.
Step 3	Probability that a sample mean ($n = 30$) exceeds 12: $Z = (12 - 10.035) / 1.835 = 1.07$, so $P = 1 - \Phi(1.07) \approx 0.142$.

Result in our analysis. The simulated standard errors were 1.829 (n = 30) and 1.023 (n = 100), and the simulated proportion above 12 was 0.1410.

Code and results: topic_04_sampling_clt

5 Confidence intervals

Theory. A point estimate is a single best guess for a parameter. The width of an interval grows with the confidence level and with the variability of the data, and shrinks as the sample size grows. Other parameters need other methods: the Wilson interval for a proportion behaves better than the simple Wald interval for small samples, the chi-square distribution gives an interval for a variance, and the Welch method gives an interval for a difference between two means without assuming equal variances.

Formulation.

Mean (t interval)	$\bar{x} \pm t^* \cdot s/\sqrt{n}$, with t^* from the t distribution with $n - 1$ df
Proportion (Wald)	$\hat{p} \pm z^* \cdot \sqrt{(\hat{p}(1-\hat{p}) / n)}$
Proportion (Wilson)	$[\, \hat{p} + z^2/(2n) \pm z \sqrt{(\, \hat{p}(1-\hat{p})/n + z^2/(4n^2)\,)}\,] / (1 + z^2/n)$
Variance	$(\, (n-1)s^2 / \chi^2\mathrm{upper}\,,\ (n-1)s^2 / \chi^2\mathrm{lower}\,)$, with $n - 1$ df
Difference in means (Welch)	$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{({s_1}^2/n_1 + {s_2}^2/n_2)}$
Sample size for a mean	$n = (z^* \sigma / E)^2$, where E is the desired margin of error

Worked example.

Given	**Courier A delivery times: $n = 40$, $\bar{x} = 32.233$, $s = 4.952$**
Standard error	$\mathrm{SE} = 4.952 / \sqrt{40} = 0.783$.
Critical value for 95% with 39 df	$t^* = 2.023$.
Margin of error	$2.023 \times 0.783 = 1.584$, so the interval is $32.233 \pm 1.584 = (30.649, 33.817)$.
Proportion satisfied	$\hat{p} = 140/200 = 0.70$, $\mathrm{SE} = \sqrt{(0.70 \times 0.30 / 200)} = 0.0324$, and $0.70 \pm 1.96 \times 0.0324$ gives (0.636, 0.764).

Result in our analysis. The Python intervals are (30.649, 33.816) for the mean and (0.6365, 0.7635) for the Wald proportion interval.

Code and results: topic_05_confidence_intervals

6 Hypothesis testing

Theory. A hypothesis test weighs evidence against a null hypothesis H_0 (usually 'no effect' or 'no difference') in favour of an alternative H_1. A test statistic measures how far the data are from what H_0 predicts. We reject

H_0 when the p-value is below the significance level α (often 0.05), which is also the probability of a Type I error, rejecting a true H_0. A Type II error is failing to reject a false H_0, and power $(1 - \beta)$ is the probability of detecting a real effect. The choice of test depends on the data: the one-sample t-test compares a mean with a target, the two-sample (Welch) t-test compares two independent means, the paired t-test compares two measurements on the same subjects, and z-tests compare proportions.

Formulation.

One-sample t	$t = (\bar{x} - \mu_0) / (s / \sqrt{n}),\ df = n - 1$
Two-sample Welch t	$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$
Paired t	$t = \bar{d} / (s_d / \sqrt{n})$, where d is the within-pair difference
One-proportion z	$z = (\hat{p} - p_0) / \sqrt{(p_0(1 - p_0) / n)}$
Two-proportion z	$z = (\hat{p}_1 - \hat{p}_2) / \sqrt{(\hat{p}(1 - \hat{p})(1/n_1 + 1/n_2))}$, $\hat{p}$ = pooled proportion
Effect size (Cohen's d)	$d = (\bar{x}_1 - \bar{x}_2) / s_pooled$
Decision rule	Reject H_0 if p-value $< \alpha$

Worked example.

Given	**Are bottles filled to 500 ml? ($n = 35, \bar{x} = 497.263, s = 4.986$)**
Hypotheses	H_0: $\mu = 500$ against H_1: $\mu \neq 500$, with $\alpha = 0.05$.
Standard error	$4.986 / \sqrt{35} = 0.843$.
Test statistic	$t = (497.263 - 500) / 0.843 = -3.248$ with 34 df.
Step 4	The critical values are ± 2.032 and $\lvert -3.248 \rvert$ is larger, so we reject H_0 ($p = 0.0026$): the mean fill is below 500 ml.
One-proportion check	62 conversions in 500 visitors give $\hat{p} = 0.124$ and $z = (0.124 - 0.10) / \sqrt{(0.10 \times 0.90 / 500)} = 1.789$ ($p = 0.0736$, not significant).

Result in our analysis. The Python output gives $t = -3.248$, $p = 0.0026$ and $z = 1.789$, $p = 0.0736$.

Code and results: topic_06_hypothesis_testing

7 Correlation and covariance

Theory. Covariance measures whether two variables move together: it is positive when large values of one go with large values of the other, and negative when large values of one go with small values of the other. Pearson's correlation coefficient divides the covariance by the two standard deviations, giving a unit-free value between −1 and 1 that measures the strength of the linear relationship. A correlation near zero means no linear association, not no relationship: a U-shaped pattern can have zero correlation.

Formulation.

Sample covariance	$\text{cov}(X, Y) = \Sigma\ (x_i - \bar{x})(y_i - \bar{y}) / (n - 1)$
Pearson correlation	$r = \text{cov}(X, Y) / (s_x\ s_y),\ -1 \leq r \leq 1$
Coefficient of determination	r^2 = share of variation in Y linearly associated with X
Covariance matrix	$S = Xc'Xc / (n - 1)$, where Xc is the centred data matrix
Spearman's rho	Pearson correlation of the ranks; $\rho = 1 - 6\Sigma d^2 / (n(n^2 - 1))$ with no ties
Kendall's tau	$\tau = (C - D) / (n(n - 1)/2)$, C = concordant pairs, D = discordant pairs
Test of r	$t = r\sqrt{(n - 2)} / \sqrt{(1 - r^2)},\ df = n - 2$

Worked example.

Given	**Study hours and exam score ($n = 200$)**
From the covariance matrix	$\text{cov}(\text{study}, \text{exam}) = 11.442$, $\text{var}(\text{study}) = 1.750$ and $\text{var}(\text{exam}) = 116.063$.
Correlation	$r = 11.442 / \sqrt{(1.750 \times 116.063)} = 11.442 / 14.252 = 0.803$.
Coefficient of determination	$r^2 = 0.803^2 = 0.645$, so 64.5% of the variation in exam score is linearly associated with study hours.

Fisher interval	$\operatorname{arctanh}(0.803) = 1.107$ and $1/\sqrt{197} = 0.0712$, so the interval on this scale is 1.107 ± 0.140. Back-transforming with tanh gives (0.747, 0.847).

Result in our analysis. The Python output gives r = 0.803 with 95% interval (0.747, 0.847).

Code and results: topic_07_correlation_covariance

8 Linear regression and least squares

Theory. Linear regression models a response variable as a linear function of one or more predictors plus random error. Each coefficient is the expected change in the response for a one-unit increase in that predictor, holding the others fixed. The model assumes a linear relationship, independent errors, constant variance and approximately normal errors.

Formulation.

Simple model	$y = \beta_0 + \beta_1 x + \varepsilon$
Least squares criterion	Minimise $SSE = \Sigma\,(y_i - \hat{y}_i)^2$
Simple regression estimates	$b_1 = Sxy \,/\, Sxx,\ b_0 = \bar{y} - b_1\bar{x},$ where $Sxy = \Sigma(x-\bar{x})(y-\bar{y})$ and $Sxx = \Sigma(x-\bar{x})^2$
Matrix solution	$\hat{\beta} = (X'X)^{-1}\,X'y$
Coefficient of determination	$R^2 = 1 - SSE \,/\, SST;$ adjusted $R^2 = 1 - (1-R^2)(n-1) \,/\, (n-p-1)$
Standard error of the slope	$SE(b_1) = s \,/\, \sqrt{Sxx},$ with $s^2 = SSE \,/\, (n-2)$
t-test for a coefficient	$t = b \,/\, SE(b),\ df = n - p - 1$

Worked example.

Given	**Data: x = 1, 2, 3, 4, 5 and y = 2, 4, 5, 4, 5**
Means	$\bar{x} = 3$ and $\bar{y} = 4$.
Step 2	$Sxy = (-2)(-2) + (-1)(0) + (0)(1) + (1)(0) + (2)(1) = 6$, and $Sxx = 4 + 1 + 0 + 1 + 4 = 10$.
Slope	$b_1 = 6 \,/\, 10 = 0.6$. Intercept: $b_0 = 4 - 0.6 \times 3 = 2.2$. Fitted line: $\hat{y} = 2.2 + 0.6x$.
Fitted values	2.8, 3.4, 4.0, 4.6, 5.2. Residuals: -0.8, 0.6, 1.0, -0.6, -0.2, so $SSE = 2.4$. $SST = 6$.
Coefficient of determination	$R^2 = 1 - 2.4 \,/\, 6 = 0.60$.

Result in our analysis. In the student-score analysis the same formulas, computed by hand in Python, gave an intercept of 38.511 and a slope of 6.538, identical to statsmodels.

Code and results: topic_08_linear_regression

9 Matrix algebra in statistics

Theory. Matrices store data and parameters in a compact form, and most statistical formulas are matrix operations. The determinant of a square matrix is nonzero exactly when the matrix has an inverse. Key statistical results are matrix expressions: the covariance matrix is $Xc'Xc/(n-1)$, the least squares coefficients solve the normal equations $(X'X)\beta = X'y$, the hat matrix H projects y onto the fitted values, and the Mahalanobis distance uses the inverse covariance matrix.

Formulation.

Addition	$(A + B)_{ij} = a_{ij} + b_{ij}$
Multiplication	$(AB)_{ij} = \Sigma_k\, a_{ik}\, b_{kj};$ in general $AB \neq BA$
Transpose rule	$(AB)' = B'A'$
Determinant (2 × 2)	$\det[[a, b], [c, d]] = ad - bc;\ \det(AB) = \det(A)\det(B)$
Inverse (2 × 2)	$A^{-1} = (1 \,/\, (ad - bc))\,[[d, -b], [-c, a]];\ AA^{-1} = I;\ (AB)^{-1} = B^{-1}A^{-1}$
Linear system	$Ax = b$ gives $x = A^{-1}b$
Normal equations	$(X'X)\,\hat{\beta} = X'y,$ so $\hat{\beta} = (X'X)^{-1}\,X'y$

Worked example.

Given	**Product, inverse and a linear system**
Product	$[[1, 2], [3, 4]] \times [[0, 1], [1, 0]] = [[1\times0 + 2\times1, 1\times1 + 2\times0], [3\times0 + 4\times1, 3\times1 + 4\times0]] = [[2, 1], [4, 3]]$.
Inverse	for $M = [[4, 7], [2, 6]]$, $\det(M) = 4\times6 - 7\times2 = 10$, so $M^{-1} = (1/10)\ [[6, -7], [-2, 4]] = [[0.6, -0.7], [-0.2, 0.4]]$.
System	$2x + y = 5$ and $x - y = 1$. Adding the equations gives $3x = 6$, so $x = 2$ and $y = 1$.
Check with matrices	$A = [[2, 1], [1, -1]]$ has $\det = -3$ and $x = A^{-1}b = (2, 1)$.

Result in our analysis. numpy returns the same inverse and the same solution (x = 2, y = 1). The normal equations reproduce the Topic 8 regression coefficients exactly.

Code and results: topic_09_matrix_algebra

10 Eigenvalues, eigenvectors and PCA

Theory. An eigenvector of a square matrix A is a direction that A only stretches, without rotating: $Av = \lambda v$. Principal component analysis (PCA) finds new, uncorrelated axes that capture as much variance as possible. Variables measured on different scales should be standardised first, equivalent to using the correlation matrix, otherwise the variable with the largest variance dominates.

Formulation.

Eigenvalue equation	$A\ v = \lambda\ v$
Characteristic equation	$\det(A - \lambda I) = 0;$ for 2×2: $\lambda^2 - \text{trace}(A)\ \lambda + \det(A) = 0$
Sum and product	$\Sigma\ \lambda_i = \text{trace}(A);\ \Pi\ \lambda_i = \det(A)$
Spectral decomposition	$R = V\ \Lambda\ V'$, with orthonormal eigenvector matrix V
Component scores	Scores $= Z\ V$, where Z is the standardised data
Proportion of variance	$\lambda_j\ /\ \Sigma\ \lambda$
Loadings	Loading of variable i on component $j = v_{ij}\ \sqrt{\lambda_j}$

Worked example.

Given	**Matrix $S = [[4, 2], [2, 3]]$**
Step 1	Trace $= 7$ and determinant $= 4\times3 - 2\times2 = 8$, so the characteristic equation is $\lambda^2 - 7\lambda + 8 = 0$.
Eigenvalues	$\lambda = (7 \pm \sqrt{17})\ /\ 2 = (7 \pm 4.123)\ /\ 2$, giving $\lambda_1 = 5.5616$ and $\lambda_2 = 1.4384$.
Eigenvector for λ_1	$(4 - 5.5616)v_1 + 2v_2 = 0$, so $v_2 = 0.7808\ v_1$. Normalising to length 1 gives $(0.7882, 0.6154)$.
Checks	$5.5616 + 1.4384 = 7$ (the trace) and $5.5616 \times 1.4384 = 8$ (the determinant).

Result in our analysis. numpy gives the same eigenvalues and eigenvectors. In the 7-subject assessment data, two components explained 71.6% of the variance.

Code and results: topic_10_eigen_pca

11 Numerical optimization

Theory. Optimization finds the input that minimises (or maximises) an objective function. Gradient-based methods use the slope of the function: gradient descent moves downhill with a step size, Newton's method uses the curvature (second derivative) and converges very fast near the optimum, and quasi-Newton methods such as BFGS approximate the curvature. Local methods find the nearest minimum, so functions with many local minima need a global method such as differential evolution, or many starting points.

Formulation.

Optimality conditions	$f'(x^*) = 0$ (stationary point); $f''(x^*) > 0$ (minimum)
Gradient descent	$x_{k+1} = x_k - \alpha\ \nabla f(x_k)$
Newton's method	$x_{k+1} = x_k - f'(x_k)\ /\ f''(x_k)$; in several variables, $x_{k+1} = x_k - H^{-1}\ \nabla f$
Step-size limit	For $f = \frac{1}{2}\ \Sigma\ c_i\ x_i^2$ gradient descent converges only if $\alpha < 2\ /\ \max(c_i)$

Maximum likelihood	$\hat{\theta} = \text{argmin of } -\ln L(\theta)$
Minimum-variance portfolio	Minimise $w'\Sigma w$ subject to $\Sigma w_i = 1$; without other constraints $w = \Sigma^{-1}1 / (1'\Sigma^{-1}1)$
Sharpe ratio	$(w'\mu - r_f) / \sqrt{(w'\Sigma w)}$

Worked example.

Given	**Minimise $f(x) = x^4 - 3x^3 + 2$**
Derivative	$f'(x) = 4x^3 - 9x^2 = x^2(4x - 9) = 0$, so $x = 0$ or $x = 9/4 = 2.25$.
Second derivative	$f''(x) = 12x^2 - 18x$. At $x = 2.25$, $f'' = 60.75 - 40.5 = 20.25 > 0$, so it is a minimum with $f(2.25) = -6.543$. At $x = 0$, $f'' = 0$, so it is not a minimum.
Newton from $x_0 = 3$	$f'(3) = 27$ and $f''(3) = 54$, so $x_1 = 3 - 27/54 = 2.5$, which is already close to 2.25.
Step 4	Gradient descent on $f = x^2 + 10y^2$ has curvatures 2 and 20. The y-update multiplies by $(1 - 20\alpha)$, which has magnitude below 1 only for $\alpha < 0.1$, so a step of 0.11 diverges.

Result in our analysis. Newton's iterations in Python were 2.5, 2.2917, 2.2515, 2.2500019 and 2.25 (5 iterations). With step 0.11 the gradient descent diverged, as predicted.

Code and results: topic_11_numerical_optimization

12 Random numbers and Monte Carlo methods

Theory. Computers generate pseudo-random numbers with deterministic algorithms that behave like random draws. Uniform numbers can be turned into other distributions. Monte Carlo methods approximate quantities such as probabilities and integrals by averaging over many random draws.

Formulation.

Linear congruential generator	$x_{k+1} = (a\,x_k + c) \bmod m;\ u_k = x_k / m$
Inverse transform	$X = F^{-1}(U)$; exponential: $X = -\ln(1 - U) / \lambda$
Box-Muller	$Z_1 = \sqrt{(-2 \ln U_1)} \cos(2\pi U_2),\ Z_2 = \sqrt{(-2 \ln U_1)} \sin(2\pi U_2)$
Acceptance-rejection	Accept a proposal x if $U \le f(x) / (M\,g(x))$; acceptance rate $= 1/M$
Monte Carlo estimate	$\hat{\theta} = (1/N)\,\Sigma\,h(X_i);\ SE = \sigma / \sqrt{N}$
Estimating pi	$\hat{\pi} = 4 \times$ (points inside the quarter circle) $/ N;\ SE = 4\sqrt{(p(1-p)/N)},\ p = \pi/4$
Antithetic variates	$\hat{\theta} = (1/N)\,\Sigma\,[h(U_i) + h(1 - U_i)] / 2$

Worked example.

Given	**A tiny generator, a transformed uniform and a Monte Carlo error**
Step 1	Generator $x \to (5x + 3) \bmod 16$ from seed 1 gives 1, 8, 11, 10, 5, 12, 15, 14, 9, 0, 3, 2, 13, 4, 7, 6 and then back to 1. It visits all 16 values, so its period is 16.
Inverse transform	for $\lambda = 0.5$ and $U = 0.5$, $X = -\ln(1 - 0.5) / 0.5 = 0.693 / 0.5 = 1.386$, the median of the exponential distribution.
Error of the pi estimate	with $N = 1{,}000{,}000$ and $p = \pi/4 = 0.7854$, $SE = 4 \times \sqrt{(0.7854 \times 0.2146 / 1{,}000{,}000)} = 4 \times 0.000411 = 0.00164$.

Result in our analysis. The Python run gave the full period of 16, and an estimate of 3.14197 for pi with a standard error of 0.00164 and a 95% interval (3.13875, 3.14519) containing pi.

Code and results: topic_12_random_monte_carlo

13 Time-series analysis

Theory. A time series is a sequence of observations ordered in time. Many methods need a stationary series, one whose mean and variance do not change over time. The autocorrelation function (ACF) measures the correlation of a series with its own past, and the partial autocorrelation function (PACF) isolates the effect of each lag.

Formulation.

Additive decomposition	$y_t = T_t + S_t + R_t$ (multiplicative: $y_t = T_t \times S_t \times R_t$)
Linear trend	$y_t = a + b\,t + \varepsilon_t$
Differencing	$\nabla y_t = y_t - y_{t-1}$; seasonal difference $\nabla_{12} y_t = y_t - y_{t-12}$
Autocorrelation	$\hat{\rho}_k = \Sigma\,(y_t - \bar{y})(y_{t+k} - \bar{y}) \,/\, \Sigma\,(y_t - \bar{y})^2$; white-noise bound $\pm 1.96 / \sqrt{n}$
AR(1) process	$y_t = \varphi\, y_{t-1} + \varepsilon_t$
ARIMA(p, d, q)	AR terms of order p, d differences, MA terms of order q; SARIMA adds seasonal terms $(P, D, Q)_m$
Ljung-Box test	$Q = n(n+2)\,\Sigma_k\, \hat{\rho}_k{}^2 / (n-k)$, compared with χ^2 (H_0: no autocorrelation)

Worked example.

Given	**Accuracy measures and differencing on small data**
Step 1	Actual values 100, 110, 120 and forecasts 98, 113, 119 give errors 2, −3, 1.
Step 2	MAE $= (2 + 3 + 1) / 3 = 2.0$. RMSE $= \sqrt{((4 + 9 + 1) / 3)} = \sqrt{4.667} = 2.16$.
Step 3	MAPE $= (2/100 + 3/110 + 1/120) / 3 \times 100 = (0.0200 + 0.0273 + 0.0083) / 3 \times 100 = 1.85\%$.
Differencing	the series 5, 8, 12, 15 has first differences 3, 4, 3, which fluctuate around a constant level, so the upward trend has been removed.
Step 5	White-noise bound for the monthly sales series with $n = 120$: $1.96 / \sqrt{120} = 0.179$.

Result in our analysis. In the Python output the bound was ±0.179, the trend slope was 2.002 per month, and Holt-Winters forecast the held-out year with MAPE 1.50%.

Code and results: topic_13_time_series

14 Statistical data visualization

Theory. A statistical graphic should show the shape of the data and the uncertainty, and the chart type should match the question. Some charts help check assumptions. Graphics can mislead.

Formulation.

Freedman-Diaconis bin width	$h = 2 \times IQR \times n\text{^}(-1/3)$
Scott's bin width	$h = 3.49 \times s \times n\text{^}(-1/3)$
Boxplot whiskers	Reach the most extreme points within $Q1 - 1.5\,IQR$ and $Q3 + 1.5\,IQR$
Empirical CDF	$F_n(x) =$ (number of $x_i \le x$) / n
Q-Q plot	Plot the ordered data against $\Phi^{-1}((i - 0.5) / n)$
Log-linear relationship	$\ln(y) = a + bx$ means each unit of x multiplies y by e^b, a $(e\text{^}b - 1) \times 100\%$ change
Lie factor	(size of effect shown in the graphic) / (size of effect in the data)

Worked example.

Given	**Choosing a bin width and spotting a misleading axis (employee salary data)**
Freedman-Diaconis for salary	$Q1 = 43{,}700$ and $Q3 = 62{,}500$ give $IQR = 18{,}800$. With $n = 400$, $n\text{^}(-1/3) = 0.1357$, so $h = 2 \times 18{,}800 \times 0.1357 \approx 5{,}100$.
Log-linear fit	$\ln(\text{salary}) = 10.598 + 0.0394 \times \text{experience}$, so each extra year multiplies salary by $e\text{^}0.0394 = 1.040$, a 4.0% increase.
Misleading axis	the highest department mean is 68,644 and the lowest is 42,763, a true ratio of $68{,}644 / 42{,}763 = 1.61$.
Step 4	On a bar chart whose axis starts at 30,000 the bars measure 38,644 and 12,763, so they look $38{,}644 / 12{,}763 = 3.03$ times as tall.

Result in our analysis. The Python output gives the same ratios (1.61 and 3.03) and a 4.0% salary increase per year of experience.

Code and results: topic_14_statistical_visualization

15 Data transformation and normalization

Theory. Variables are often measured on very different scales. Standardisation (z-scores) gives mean 0 and standard deviation 1. Min-max scaling maps values to [0, 1] but is very sensitive to outliers. Categorical variables are encoded before modelling: one-hot encoding for unordered categories and ordinal encoding for ordered ones.

Formulation.

Standardisation	$z = (x - \bar{x}) / s$
Min-max scaling	$x' = (x - \min) / (\max - \min)$
Robust scaling	$x' = (x - \text{median}) / \text{IQR}$
Max-abs scaling	$x' = x / \max\|x\|$
Log and square-root transforms	$y = \ln(x)\ (x > 0);\ y = \sqrt{x}\ (x \geq 0);\ y = \ln(1 + x)$ for counts with zeros
Box-Cox	$y(\lambda) = (x^{\wedge}\lambda - 1) / \lambda$ for $\lambda \neq 0;\ y = \ln x$ for $\lambda = 0$
Winsorising	Replace values below the p-th percentile or above the $(100 - p)$-th percentile by those limits

Worked example.

Given	**Data: 10, 20, 30, 40, 100**
Mean = 40. Deviations	$-30, -20, -10, 0, 60$. Squares sum to 5,000, so the population variance is 1,000 and sd = 31.62 (the divisor used by StandardScaler).
Z-score of 100	$(100 - 40) / 31.62 = 1.90$.
Min-max	$(x - 10) / 90$ gives 0, 0.111, 0.222, 0.333 and 1. The outlier 100 squeezes the other four values into the lower third of the range.
Robust	median = 30, Q1 = 20, Q3 = 40, IQR = 20, so the scaled value of 100 is $(100 - 30) / 20 = 3.5$, and the typical values stay near 0.
Box-Cox with λ = 0 is the log	$\ln(100) = 4.605$, $\ln(10) = 2.303$, a 10-fold range becomes a 2-fold range.

Result in our analysis. In the customer data, three extreme incomes raised the sd by 49.8% but the median by 0%, and only z-scoring gave every variable an equal 25% share of the squared distance.

Code and results: topic_15_data_transformation

16 ANOVA and experimental design

Theory. Analysis of variance (ANOVA) tests whether the means of several groups are equal. Running many pairwise t-tests inflates the chance of at least one false positive. The assumptions are independent observations, approximately normal residuals and equal variances.

Formulation.

One-way model	$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$
Sums of squares	$SST = SSB + SSW;\ SSB = \Sigma\ n_i\ (\bar{y}_i - \bar{y})^2;\ SSW = \Sigma\Sigma\ (y_{ij} - \bar{y}_i)^2$
Mean squares	$MSB = SSB / (k - 1);\ MSW = SSW / (N - k)$
F-statistic	$F = MSB / MSW$, with $(k - 1, N - k)$ degrees of freedom
Effect sizes	$\eta^2 = SSB / SST;\ \omega^2 = (SSB - (k - 1)\ MSW) / (SST + MSW);\ f = \sqrt{(\eta^2 / (1 - \eta^2))}$
Tukey HSD	$HSD = q\ \sqrt{(MSW / n)}$, q from the studentised range distribution
Bonferroni	Compare each of m p-values with α / m

Worked example.

Given	**Three groups: A = 4, 5, 6; B = 6, 7, 8; C = 8, 9, 10**
Group means	5, 7 and 9. Grand mean = 7.
Between groups	$SSB = 3(5-7)^2 + 3(7-7)^2 + 3(9-7)^2 = 12 + 0 + 12 = 24$, with $k - 1 = 2$ df, so $MSB = 12$.
Within groups	each group has deviations $-1, 0, 1$, so $SSW = 2 + 2 + 2 = 6$ with $N - k = 6$ df, so $MSW = 1$.
Step 4	$F = MSB / MSW = 12 / 1 = 12$ on (2, 6) df, and for these df $p = (1 + 2F/6)^{-3} = 5^{-3} = 0.008$, so the means differ.
Effect size	$\eta^2 = 24 / (24 + 6) = 0.80$.

Result in our analysis. In the plant-growth data, F = 9.631 on (3, 56) df gave p = 3.181e-05. In the block design the variety F rose from 0.78 (blocks ignored) to 8.05 (blocks included).

Code and results: topic_16_anova_experimental_design

17 Chi-square and categorical data analysis

Theory. Categorical data are summarised as counts in a contingency table. The chi-square statistic is reliable when expected counts are not too small (the usual guideline is at least 5). For a 2×2 table, the risk difference, relative risk and odds ratio describe the effect.

Formulation.

Chi-square statistic	$\chi^2 = \Sigma\,(O - E)^2 / E$
Goodness of fit	$df = k - 1$ (categories minus 1)
Independence	$E = (\text{row total} \times \text{column total}) / N$; $df = (r-1)(c-1)$
G-test	$G = 2\,\Sigma\, O \ln(O / E)$
Cramér's V	$V = \sqrt{(\chi^2 / (N \times \min(r-1, c-1)))}$
Standardised residual	$(O - E) / \sqrt{E}$
2×2 table [[a, b], [c, d]]	Risk difference $= a/(a+b) - c/(c+d)$; Odds ratio $OR = ad / bc$; $SE(\ln OR) = \sqrt{(1/a + 1/b + 1/c + 1/d)}$

Worked example.

Given	**Drug against placebo: drug 140 recovered and 60 not; placebo 109 recovered and 91 not**
Totals	249 recovered and 151 not, out of 400. Expected recoveries per arm if there is no association: $200 \times 249 / 400 = 124.5$, and expected non-recoveries 75.5.
Chi-square	$(140 - 124.5)^2/124.5 + (60 - 75.5)^2/75.5 + (109 - 124.5)^2/124.5 + (91 - 75.5)^2/75.5 = 1.930 + 3.182 + 1.930 + 3.182 = 10.224$, with $df = 1$.
Step 3	The critical value at 5% is 3.841, so we reject independence ($p = 0.0014$).
Odds ratio	$(140 \times 91) / (60 \times 109) = 12{,}740 / 6{,}540 = 1.948$. Risk difference: $0.700 - 0.545 = 0.155$.

Result in our analysis. The Python output gives $\chi^2 = 10.224$, p = 0.0014 and an odds ratio of 1.948.

Code and results: topic_17_chi_square

18 Logistic regression

Theory. Logistic regression models a binary outcome (such as churn: yes or no). Coefficients are estimated by maximum likelihood. Predicted probabilities become class predictions through a threshold.

Formulation.

Logit model	$\ln(p / (1-p)) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k = \eta$
Probability	$p = 1 / (1 + e^{-\eta})$
Odds and odds ratio	odds $= p / (1-p)$; odds ratio for $x_j = e^{\beta_j}$
Likelihood	$L(\beta) = \Pi\, p_i^{y_i} (1-p_i)^{(1-y_i)}$; $\hat{\beta}$ maximises $\ln L$

Likelihood ratio test	$LR = 2\,(\ln L_model - \ln L_null)$, compared with χ^2 (k df)
McFadden R^2	$1 - \ln L_model / \ln L_null$
Classification metrics	$Accuracy = (TP + TN)/N$; $Precision = TP/(TP + FP)$; $Recall = TP/(TP + FN)$; $Specificity = TN/(TN + FP)$; $F1 = 2PR/(P + R)$

Worked example.

Given	**A high-risk customer, and metrics from a confusion matrix**
Step 1	A customer with 3 months of tenure, \$90 charges, 4 calls and a month-to-month contract has log-odds $\eta = 1.2881$.
Step 2	Odds = e^1.2881 = 3.626. Probability = 3.626 / (1 + 3.626) = 0.784.
Odds ratio for month-to-month	e^1.0935 = 2.985, so the odds of churn are about three times those of annual contracts.
Step 4	Test-set confusion matrix at threshold 0.5: TN = 161, FP = 38, FN = 62, TP = 39 (300 customers).
Step 5	Accuracy = 200 / 300 = 0.667. Precision = 39 / 77 = 0.506. Recall = 39 / 101 = 0.386. Specificity = 161 / 199 = 0.809.

Result in our analysis. The Python output gives probability 0.7838 for the high-risk customer, odds ratio 2.9847, and the same four metrics. Always predicting 'stayed' would score accuracy 0.663.

Code and results: topic_18_logistic_regression

19 Non-parametric statistical methods

Theory. Non-parametric methods make few assumptions about the shape of the distribution. Each common parametric test has a rank-based counterpart: Mann-Whitney U for two independent groups (instead of the t-test), Wilcoxon signed-rank and the sign test for paired data, Kruskal-Wallis for several independent groups (instead of one-way ANOVA), and Friedman for several related groups. When the data really are normal, a rank test loses only a little power (efficiency about $3/\pi = 0.955$ for Mann-Whitney), and for skewed or heavy-tailed data it can be much more powerful.

Formulation.

Mann-Whitney U	$U_1 = R_1 - n_1(n_1 + 1)/2$; $U_1 + U_2 = n_1\,n_2$, where R_1 is the rank sum of sample 1
Probability of superiority	$U / (n_1\,n_2)$; rank-biserial correlation $= 2U/(n_1\,n_2) - 1$
Wilcoxon signed-rank	W^+ = sum of ranks of $\lvert d\rvert$ for positive differences; W^- for negative; zero differences dropped
Sign test	Number of positive differences $\sim$ Binomial(m, 0.5) under H_0
Kruskal-Wallis	$H = 12 / (N(N + 1))\ \Sigma\, R_i^2 / n_i - 3(N + 1)$, compared with χ^2 (k − 1 df)
Friedman	$\chi^2_F = 12 / (n\,k\,(k + 1))\ \Sigma\, R_j^2 - 3n(k + 1)$; Kendall's $W = \chi^2_F / (n(k - 1))$
Spearman and Kendall	ρ = Pearson correlation of the ranks; $\tau = (C - D) / (n(n - 1)/2)$

Worked example.

Given	**Samples A = 12, 15, 9, 20, 17 and B = 8, 14, 11, 7, 10**
Combine and rank the 10 values	7→1, 8→2, 9→3, 10→4, 11→5, 12→6, 14→7, 15→8, 17→9, 20→10.
Ranks of A	6, 8, 3, 10, 9, so R_A = 36. Ranks of B: 2, 7, 5, 1, 4, so R_B = 19. Check: 36 + 19 = 55 = 10 × 11 / 2.
Step 3	$U_A = R_A - n(n + 1)/2 = 36 - 15 = 21$ and $U_B = 25 - 21 = 4$. The smaller value, 4, is compared with the exact table.
Step 4	The exact two-sided p-value is 0.0952, so with only 5 per group the difference is not significant at 5%.

Result in our analysis. scipy gives U = 21 and the exact p = 0.0952. On skewed data with 20 per group, the simulated power was 0.847 for Mann-Whitney against 0.454 for the t-test.

Code and results: topic_19_nonparametric

20 Bootstrap and resampling methods

Theory. The bootstrap estimates the sampling distribution of a statistic by resampling the data. This works for statistics that have no simple formula, such as the median, the trimmed mean or the coefficient of variation. The bootstrap is not magic.

Formulation.

Bootstrap standard error	$\text{SE_boot} = \sqrt{(\ \Sigma\ (\theta^*_b - \bar{\theta}^*)^2\ /\ (B-1)\ )}$
Bootstrap bias	$\text{bias} = \bar{\theta}^* - \hat{\theta}$
Percentile interval	(2.5th percentile, 97.5th percentile) of the bootstrap values
Basic interval	$(\ 2\hat{\theta} - q_{97.5}\ ,\ 2\hat{\theta} - q_{2.5}\ )$
Jackknife standard error	$\text{SE_jack} = \sqrt{(\ ((n-1)/n)\ \Sigma\ (\theta_{(i)} - \bar{\theta}_{(\cdot)})^2\ )}$
Chance a point is in a resample	$1 - (1 - 1/n)^n \to 1 - e^{-1} \approx 0.632$
Pairs bootstrap for regression	Resample whole (x_i, y_i) pairs, refit, and keep the slope

Worked example.

Given	**Incomes of 40 customers**
Step 1	Sample mean = 41,892 and sample sd = 22,138. Formula standard error: $22{,}138 / \sqrt{40} = 22{,}138 / 6.325 = 3{,}500$.
Bootstrap	draw 10,000 resamples of size 40, compute each mean, and take the standard deviation of the 10,000 means. This gives 3,514, almost the same as the formula.
Percentile interval	the 2.5th and 97.5th percentiles of the bootstrap means are 35,522 and 49,236.
Why the maximum fails	a resample misses a given point with probability $(1 - 1/50)^{50} = 0.364$, so about 63.6% of resample maxima equal the sample maximum, and the true upper limit can never appear.

Result in our analysis. The Python results are 3,514.4 (bootstrap) against 3,500.3 (formula), and a share of 0.633 for the maximum example (theory 0.636).

Code and results: topic_20_bootstrap_resampling

21 Multivariate statistical analysis

Theory. Multivariate analysis studies several variables together, using their correlations. Hotelling's T^2 compares the mean vectors of two groups, and MANOVA does so for several groups, using Wilks' lambda and related statistics. Principal component analysis reduces dimension, and clustering (k-means, hierarchical, Gaussian mixture) looks for groups without using labels.

Formulation.

Mahalanobis distance	$D^2 = (x - \mu)'\ S^{-1}\ (x - \mu),$ approximately χ^2 with p df
Hotelling's T^2	$T^2 = (n_1 n_2\ /\ (n_1 + n_2))\ (\bar{x}_1 - \bar{x}_2)'\ S_p^{-1}\ (\bar{x}_1 - \bar{x}_2)$
Conversion to F	$F = ((n_1 + n_2 - p - 1)\ /\ (p\ (n_1 + n_2 - 2)))\ T^2,$ with $(p, n_1 + n_2 - p - 1)$ df
Wilks' lambda	$\Lambda = \lvert W\rvert\ /\ \lvert T\rvert,$ where W is the within-group and T the total sums-of-squares-and-products matrix
Linear discriminant analysis	Assign x to the group with the largest score $\delta_k(x) = x'\ S_p^{-1}\ \mu_k - ½\ \mu_k'\ S_p^{-1}\ \mu_k + \ln \pi_k$
k-means	Minimise the within-cluster sum of squares $\Sigma_k\ \Sigma\ \lVert x - c_k\rVert^2$
Silhouette	$s = (b - a)\ /\ \max(a, b),$ a = mean distance within the cluster, b = to the nearest other cluster

Worked example.

Given	**Mahalanobis distance with $\Sigma = [[4, 2], [2, 3]]$ and $x - \mu = (2, 1)$**

Step 1	Determinant of $\Sigma = 4\times3 - 2\times2 = 8$, so $\Sigma^{-1} = (1/8)\ [[3, -2], [-2, 4]]$.
Step 2	$\Sigma^{-1}(x - \mu) = (1/8)\ (3\times2 - 2\times1, -2\times2 + 4\times1) = (1/8)(4, 0) = (0.5, 0)$.
Step 3	$D^2 = (x - \mu)'\ \Sigma^{-1}\ (x - \mu) = (2, 1) \cdot (0.5, 0) = 1$.
Step 4	For comparison, the ordinary squared distance is $2^2 + 1^2 = 5$. Mahalanobis distance is smaller here because the two variables are positively correlated and the point lies along the main axis of the data.
Step 5	Wilks' lambda for the three-species data: $\Lambda = \lvert W\rvert\ /\ \lvert T\rvert = 0.4557$, so about 54% of the multivariate variation is associated with species.

Result in our analysis. Python gave $\Lambda = 0.4557$ (matching the hand formula) with $p < 0.0001$, yet LDA reached only 61.7% cross-validated accuracy, because the species overlap.

Code and results: topic_21_multivariate

22 Missing-data analysis and imputation

Theory. Missing values should first be described: how much is missing, in which variables, and in what patterns. Complete-case analysis discards every row with a missing value, which loses information and can bias results. Multiple imputation creates several completed datasets, analyses each, and pools the results with Rubin's rules, so that the standard errors include the uncertainty caused by the missing values.

Formulation.

Missingness indicator	$R_i = 1$ if the value is missing, 0 if observed
MCAR	$P(R \mid Y, X) = P(R)$
MAR	$P(R \mid Y, X) = P(R \mid X_observed)$
MNAR	$P(R \mid Y, X)$ depends on the missing Y itself
Pooled estimate	$\bar{Q} = (1/m)\ \Sigma\ Q_i$
Within and between variance	$\bar{W} = (1/m)\ \Sigma\ U_i;\ \ B = \Sigma\ (Q_i - \bar{Q})^2\ /\ (m - 1)$
Total variance (Rubin's rules)	$T = \bar{W} + (1 + 1/m)\ B;\ \ \text{pooled SE} = \sqrt{T}$

Worked example.

Given	**Rubin's rules with m = 3 imputed datasets**
Step 1	Suppose the three completed datasets give estimates $Q = 10, 12, 11$, each with variance $U = 1$.
Pooled estimate	$\bar{Q} = (10 + 12 + 11)\ /\ 3 = 11$.
Within-imputation variance	$\bar{W} = 1$. Between-imputation variance: $B = ((-1)^2 + 1^2 + 0^2)\ /\ 2 = 1$.
Total variance	$T = 1 + (1 + 1/3) \times 1 = 2.333$, so the pooled $SE = \sqrt{2.333} = 1.528$, larger than the SE of any single dataset (1), as it should be.
Fraction of missing information	$\lambda = (4/3)\ /\ 2.333 = 0.571$.

Result in our analysis. In the analysis, 20 imputations gave a pooled SE of 0.4317 for mean income. For health scores (MNAR) every method overestimated the mean by 1.4 to 2.1 points.

Code and results: topic_22_missing_data

23 Statistical reporting and reproducible analysis

Theory. A reproducible report lets anyone with the same data and code regenerate every table, figure and number. Reproducibility needs more than the code. A good statistical report states the question, describes the data and any missing values, shows a baseline table, reports effects with confidence intervals and adjusted p-values, checks the model assumptions, runs sensitivity analyses, and states limitations honestly.

Formulation.

Adjusted treatment effect	$\text{change} = \beta_0 + \beta_A \cdot \text{DrugA} + \beta_B \cdot \text{DrugB} + \beta_1 \cdot \text{baseline} + \beta_2 \cdot \text{age} + \beta_3 \cdot \text{female} + \varepsilon$
Confidence interval	$\text{estimate} \pm t^* \times \text{SE}$
Cohen's d	$d = (\bar{x}_1 - \bar{x}_2) \,/\, s_pooled$
Bonferroni adjustment	$p_adj = \min(1, m \times p)$
Holm adjustment	Order the p-values $p(1) \le \ldots \le p(m)$; $p_adj(i) = \max$ over $j \le i$ of $\min(1, (m - j + 1)\, p(j))$
Robust standard errors	HC3 sandwich estimator, used as a sensitivity analysis
Data fingerprint	SHA-256 hash of the data file; identical data give an identical hash

Worked example.

Given	**Holm adjustment for two comparisons with control**
Unadjusted p-values	Drug A vs Control $p = 0.00326$ and Drug B vs Control $p = 1.02 \times 10^{-9}$. There are $m = 2$ comparisons.
Order them	the smallest (Drug B) is multiplied by 2, giving 2.04×10^{-9}. The next (Drug A) is multiplied by 1, giving 0.00326 (and it cannot be smaller than the previous adjusted value).
Step 3	Both adjusted p-values are below 0.05, so both drugs differ significantly from control.
Reproducibility check	regenerate the data from seed 2024 and compare hashes. The two SHA-256 values matched, starting c1f7458ea07942b8.

Result in our analysis. The notebook reports an adjusted effect of −9.63 mmHg for Drug B vs Control (95% interval −12.60 to −6.66) and −4.38 mmHg for Drug A vs Control.

Code and results: topic_23_reproducible_reporting

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